Effect of Renal Failure on the Pharmacokinetics of Acyclovir

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To determine the effect of renal failure on the pharmacokinetics of acyclovir in patients with end-stage renal disease (ESRD), we studied six anuric subjects on chronic hemodialysis. Each subject received a single one-hour intravenous infusion of acyclovir (2.5 mg/kg). We compared these anuric subjects with 13 subjects with normal renal function (NRF) who had received a single dose of acyclovir (2.5 to 15 mg/kg) in an identical fashion. The kinetics were well-described by a two-compartment open model. The mean terminal plasma half-life of acyclovir in subjects with ESRD was 19.5 ± 5.9 hours (mean ± SD) compared with 2.9 ± 0.8 hours in our subjects with NRF. In subjects with renal failure the mean (± SD) peak, eight- and 24-hour plasma acyclovir concentrations were 37.5 ± 23.3, 10.3 ± 2.9, and 6.4 ± 2.4, μM respectively.

Forty-eight hours after the start of acyclovir infusion, the subjects were hemodialyzed for six hours. The pre- and post-hemodialysis acyclovir plasma levels were 2.74 ± 1.38 and 1.11 ± 0.60 μM, respectively. The total body clearance of acyclovir (28.6 ± 9.5 ml/min/1.73 m^2) in ESRD was found to be approximately 10 percent of that previously seen in subjects with normal renal function (307 ± 98.4 mg/min/1.73 m^2). The volume of distribution at steady state was significantly less in the subjects with ESRD than in subjects with NRF. Acyclovir was readily hemodialyzable with an extraction coefficient of 0.45 ± 0.12 and a fourfold enhancement in the elimination of acyclovir during dialysis. Suggestions for acyclovir dosage modifications for patients with ESRD are provided.

Acyclovir* (9-(2-hydroxyethoxymethyl)guanine, acycloguanosine) is currently being investigated as a therapeutic agent against herpes virus infections. This compound has selective activity in cells infected with herpes virus by virtue of its specific phosphorylation by the viral-coded deoxynucleoside kinase and the selectivity of acyclovir triphosphate as both a substrate and an inhibitor of the herpes virus DNA polymerase [1,2].

Acyclovir has been shown to be efficacious *in vitro* and *in vivo* in preclinical studies [3–8]. In clinical trials in humans, acyclovir was found to be remarkably effective in preventing and treating infections caused by both herpes simplex virus and varicella zoster virus [9–14].

The pharmacokinetics of acyclovir in subjects with normal renal function has been extensively studied [15–19]. From these studies, we know that the terminal plasma half-life of acyclovir is approximately three hours, that the renal clearance of acyclovir after a single one-

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* Zovirax®, Burroughs Wellcome.

hour infusion is two to three times greater than the creatinine clearance, and that more than 80 percent of ^{14}C-acyclovir is recovered from the urine, with only 8.5 to 14 percent of dose eliminated as 9-carboxymethoxymethylguanine [17], the only significant metabolite that has been isolated to date. This suggests that acyclovir's major route of elimination is the kidneys, and involves renal tubular secretion in addition to glomerular filtration.

One would expect a drug with the pharmacokinetic profile of acyclovir to be influenced drastically by significant changes in renal function. In addition, patients with severe renal insufficiency are at increased risk of serious herpes virus infections and would therefore be likely candidates for acyclovir therapy. To determine the pharmacokinetics of acyclovir in the absence of renal function and the elimination of acyclovir during hemodialysis, acyclovir was administered to subjects with end-stage renal disease.

MATERIALS AND METHODS

Patient Selection. Six patients (four men and two women) with a mean age of 46 years (range 30 to 72 years) being treated at The Johns Hopkins Dialysis Unit were selected for this study. All had end-stage renal failure and four were surgically anephric. All were anuric. Their mean (range) weight, height, and body surface area were 57.5 kg (48 to 74 kg), 173 cm (163 to 185 cm), and 1.66 m^2 (1.51 to 1.9 m^2), respectively. Informed consent was obtained from each subject before entering the study.

Drug Administration, Sample Collection, and Acyclovir Assay. The morning after a scheduled hemodialysis, each patient received a single intravenous infusion of 2.5 mg/kg acyclovir delivered over one hour. Plasma concentrations were documented for 48 hours from the start of infusion. The subjects then underwent hemodialysis for six hours where paired arterial and venous acyclovir concentrations were followed hourly. A post-dialysis acyclovir level was obtained 15 to 90 minutes after dialysis.

Plasma concentrations of acyclovir were determined using a sensitive and specific radioimmunoassay [20]. The lower limit of detection of this assay is 0.05 μM.

Pharmacokinetic and Statistical Analysis. The data were analyzed as previously described [18,21] using a two-compartment model modified to incorporate hemodialysis by including a second elimination route and rate constant (k_{dial}) from the first compartment parallel to the usual one described by the first order rate constant of elimination (k_{el}).

Data were analyzed using the Student's t test to compare the means of two populations (end-stage renal disease versus normal renal function) and the means of a parameter calculated in the inter-dialyses period versus during hemodialysis. Differences were considered statistically significant for probability values less than 0.05.

RESULTS

The acyclovir plasma concentration–time profiles were well-described by a two-compartment open model both after the administration of acyclovir and during hemodialysis. After a one-hour infusion, the subjects' individual plasma acyclovir concentrations at one, two, eight, 12, and 24 hours and pre- and post-dialysis were recorded (**Table I**). During hemodialysis, elimination of acyclovir was more rapid. In every subject, the concentration of acyclovir rose slightly after hemodialysis. This is presumably due to the reestablishment of a new equilibrium with the peripheral compartment following dialysis.

Figure 1 shows the mean (± SD) plasma acyclovir concentrations of acyclovir (solid line) in subjects with end-stage renal disease [21]. For comparison the dotted line represents the plasma concentration of acyclovir from three subjects with normal renal function from a previous study where acyclovir was given in an identical fashion [18]. In the subjects with renal failure, the peak acyclovir concentration was 37.5 μM, compared with 20 μM in the subjects with normal renal function. At eight hours the mean acyclovir concentration was 10.3 μM, more than five times greater than the concentration of acyclovir in subjects with normal renal function. Acyclovir persisted in the plasma of the subjects with renal insufficiency; its concentration was greater than 1 μM in every subject at 48 hours. The

TABLE I Plasma Acyclovir Concentration (μM) After One-Hour Infusion (2.5 mg/kg) in Subjects with Renal Failure

Subject Number	1 hour	2 hours	8 hours	12 hours	24 hours	48 hours*	54.5 hours†
1	32.6	23.8	16.1	13.5	9.1	4.4	1.8
2	36.1	16.1	8.5	7.3	4.6	1.6	0.65
3	24.9	14.9	8.5	7.3	4.9	2.15	0.79
4	22.8	18.2	8.9	6.5	3.4	1.0	0.35
5	24.9	14.9	9.3	10.5	8.2	3.2	1.4
6	83.9	17.5	10.3	8.0	8.2	4.1	1.7
Mean	37.5	17.6	10.3	8.9	6.4	2.74	1.12
± SD	23.3	3.3	2.9	2.7	2.4	1.38	0.60

* Represents pre-dialysis acyclovir plasma concentration.
† Represents post-dialysis acyclovir concentration.
1 μM = 0.225 μg/ml

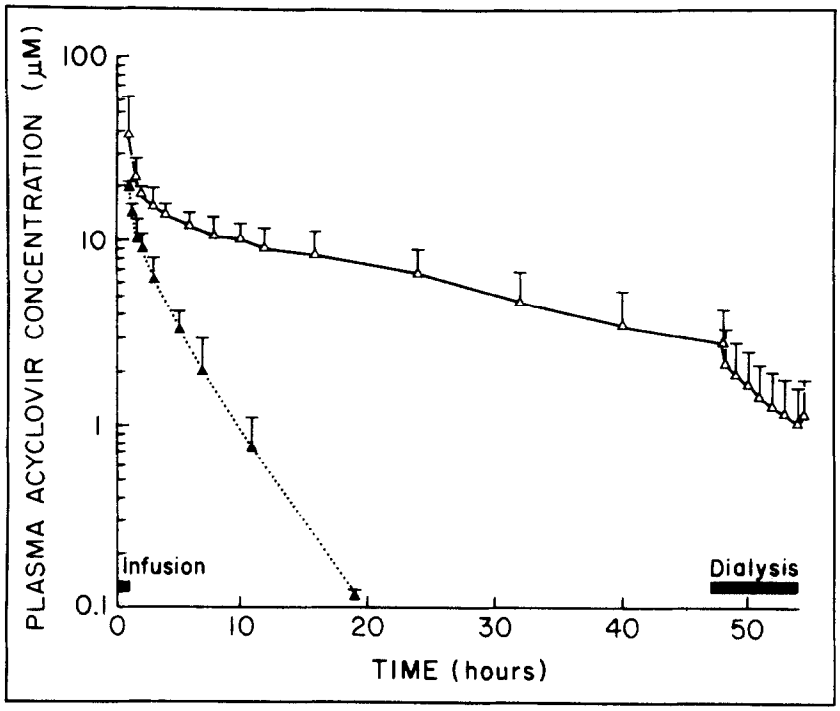


Figure 1. *Comparison of the mean (± SD) plasma acyclovir concentration versus time profiles in six subjects with end-stage renal disease (Δ—Δ) and three subjects with normal renal function (▲.....▲) after receiving a single one-hour infusion of acyclovir (2.5 mg/kg).*

mean concentration before dialysis was 2.74 μ*M*. In comparison, in subjects without renal impairment, acyclovir levels fell below 1 μ*M* by 11 hours and were undetectable (<0.05 μ*M*) at 36 hours.

Table II compares the pharmacokinetic parameters in functionally anephric subjects [21] with those from subjects with normal renal function [18]. In patients with end-stage renal disease, the terminal plasma half-life was found to be 19.5 hours. The half-lives ranged from 13 to 29 hours. This is a sevenfold increase over the terminal plasma half-life seen in our subjects with normal renal function. The apparent volume of distribution of the central compartment in subjects with ESRD was 15.3 liters/1.73 m^2 but this value was highly variable, ranging from 3.6 to 25.5 liters/1.73 m^2. The apparent volume of distribution at steady state ($V_{d_{ss}}$) was 41.2 liters/1.73 m^2. This was significantly different when compared with the value of 49.6 liters/1.73 m^2 in subjects with normal renal function. The volume of the central compartment was about 40 percent of the $V_{d_{ss}}$. The total body clearance of acyclovir in the absence of renal function was approximately 28 ml/min/1.73 m^2. This value is only about 10 percent of that observed in subjects with normal glomerular function.

During hemodialysis, the mean (±SD) terminal dialysis half-life was 5.73 ± 0.85 hours. This is significantly different from the mean terminal half-life of acyclovir in the inter-dialyses period: 19.5 ± 5.85 hours. The acyclovir dialysis clearance was found to be 82 ml/min. Therefore, hemodialysis enhanced the clearance of acyclovir so that the overall elimination was approxi-

TABLE II **Pharmacokinetic Parameters in Patients with End-Stage Renal Disease (ESRD) Compared with Patients with Normal Renal Function (NRF)**

Parameter	ESRD (n = 6) Mean ± SD	NRF (n = 13) Mean ± SD	P
Terminal $T_{1/2}$ (hr)	19.5 ± 5.9	2.9 ± 0.8	<0.001
V_1 (liters/1.73 m^2)	15.3 ± 8.1	20.8 ± 7.5	N.S.
$V_{d_{ss}}$ (liters/1.73 m^2)	41.2 ± 2.3	49.6 ± 8.7	<0.001
$V_1/V_{d_{ss}} \times 100$ (%)	37 ± 19.9	42 ± 11.4	N.S.
Cl_{tot} (ml/min/1.73 m^2)	28.6 ± 9.5	307 ± 98	<0.001
k_{el} (liters/hr)	0.15 ± 0.09	0.92 ± 0.25	<0.001

$T_{1/2}$ = half-life; V_1 = volume of distribution of central compartment; $V_{d_{ss}}$ = volume of distribution at steady state; Cl_{tot} = total body clearance of acyclovir; k_{el} = apparent first order rate constant of elimination; N.S. = not statistically significant.

TABLE III Acyclovir Dosing in Patients with End-Stage Renal Disease

	Changing Dosage	Changing Interval
Loading dose	37% of the standard dosage* (93–185 mg/m^2)	Full standard dose (250–500 mg/m^2)
Maintenance dose	14% of the standard dosage every eight hours (35–70 mg/m^2)	Full standard dose every 48 hrs (250–500 mg/m^2)
Post-dialysis	60–100% loading dose	60–100% standard dose
Potential advantages	Minimizes fluctuations between peaks and troughs	Less frequent administration

* Standard acyclovir dosage (patients with normal renal function) = 250–500 mg/m^2 initially and every eight hours.

mately four times greater than in the absence of dialysis. The coefficient of extraction of acyclovir was found to be 0.45 ± 0.12. After hemodialysis, the plasma concentration was 60 percent less than the pre-dialysis concentration of acyclovir. A more extensive analysis of the influence of hemodialysis on the pharmacokinetics of acyclovir in this group of subjects is presented in this volume [22]. The plasma acyclovir concentration determinations and the pharmacokinetic analyses were independently performed at both Wellcome Research Laboratories and The Johns Hopkins School of Medicine. Therefore they vary slightly between papers because of interassay variability and minor differences in technique.

COMMENTS

In the absence of renal clearance, the elimination of acyclovir is markedly reduced; however, it is slowly cleared by nonrenal routes of elimination. The fate of acyclovir and its metabolites in anuric subjects has not been established. Acyclovir probably undergoes significant biotransformation. While fecal and respiratory elimination were not studied, a previously reported radioactive study did not find these routes of elimination to contribute significantly in subjects with normal renal function [17]. In addition, acyclovir is readily hemodialyzable. This is predictable in view of acyclovir's low molecular weight (225), its relative solubility in water (maximum solubility = 1.25 mg/ml or 5556 μM), and its low plasma protein binding (9 to 22 percent) [17].

When acyclovir is administered to patients with end-stage renal failure, dosage modification (**Table III**) is necessary to maintain acyclovir concentrations at an effective level without concomitant accumulation.

The steady-state concentrations of acyclovir will depend on both the maintenance dosage and the interval chosen; modifications can be achieved by changing either the dosage or the dosage interval. An example of changing the dosage would be to give an initial loading dose allowing one to reach a therapeutic level and then to give a reduced maintenance dose every eight hours. Computer simulations using the two-compartment model indicate that a loading dose of 37 percent and an eight-hourly maintenance dose of 14 percent of the dosage recommended for a similar patient with normal renal function should produce and maintain similar mean plasma concentrations. An advantage of this schedule is that it minimizes the fluctuations between peak and trough concentrations. However, this advantage is only theoretic since neither toxic nor therapeutic concentrations have been established.

An example of changing the dosage interval would be to administer a full standard dose of acyclovir every second or third half-life. In the anuric patient this would be about every 48 hours. An advantage of changing the interval is that the drug is administered less frequently, which may be useful if venous access is a problem. However, this regimen results in higher peak and lower trough plasma concentrations and, if the critical acyclovir concentration necessary to inhibit viral replication is greater than the trough values, may result in long periods with suboptimal concentrations. There is no evidence that one schedule would be more efficacious or less toxic than the other. Both techniques should result in similar mean plasma concentrations for anuric patients.

During hemodialysis about 60 percent of the drug in the body will be removed and should be replaced. Depending on when dialysis occurs relative to the preceding dose, this will be 60 to 100 percent of the full loading dose. For example, if dialysis is performed immediately after a dose of acyclovir, it would be rational to give 60 percent of a loading dose after dialysis. If dialysis is performed at the end of a dosing interval (when the drug present in the body is already reduced) 100 percent of a loading dose should be given.

The data from this study should serve as a foundation for the rational design of dosing for patients with end-stage renal disease in the intervals between dialysis and after dialysis.

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